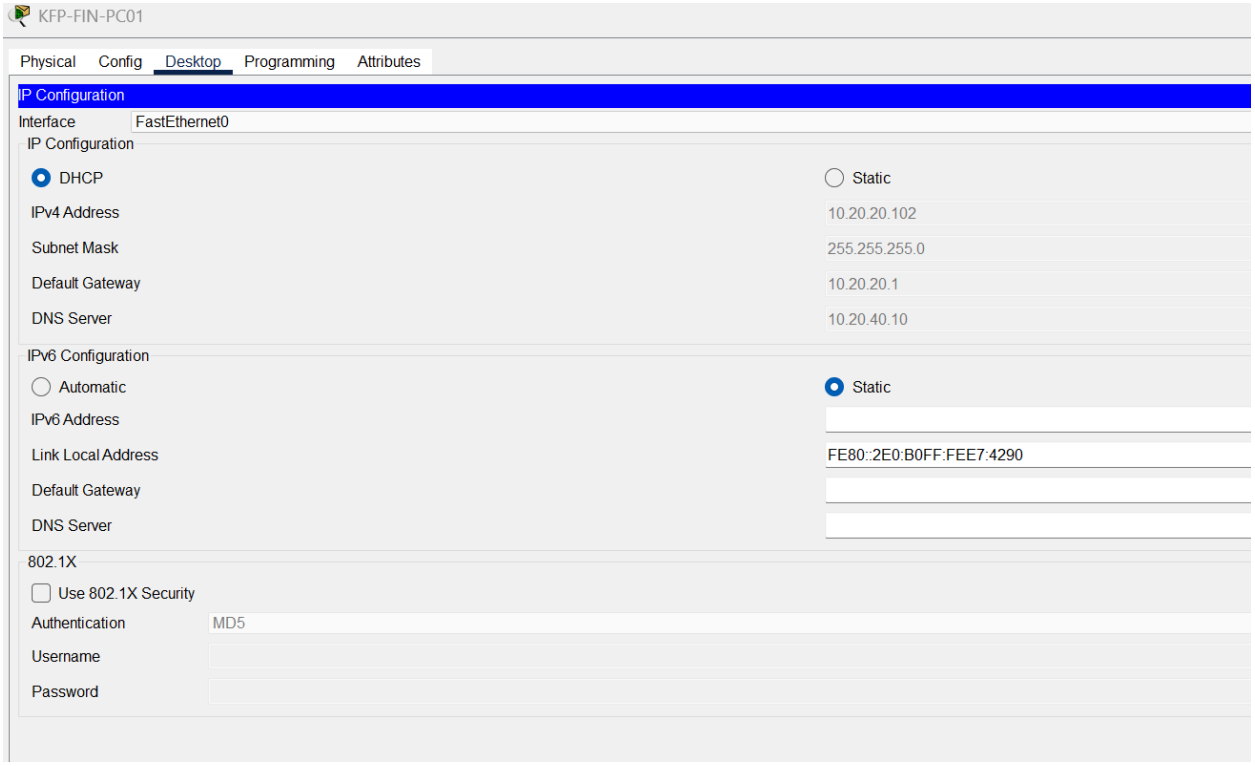
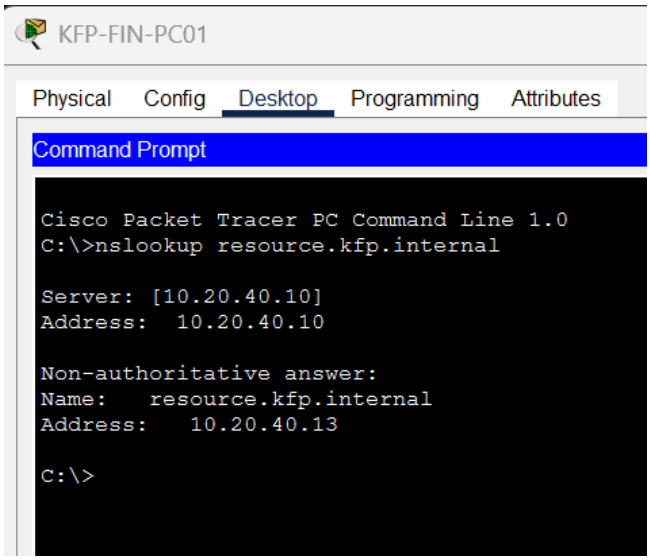
US-19 – ACL and Network-Service Integration Verification

DHCP Verification



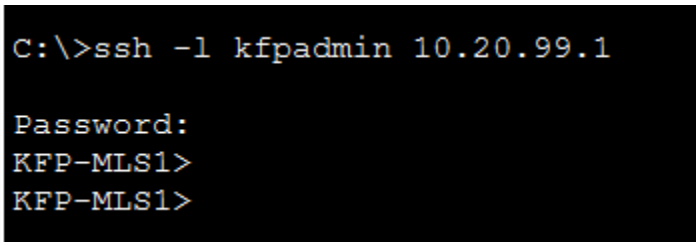
Result: Pass - Finance PC successfully received a DHCP address while the ACL was active

DNS Verification



Result: Pass - Finance PC successfully resolved resource.kfp.internal to 10.20.40.13.

RADIUS and SSH Verification



Result: Pass - IT/Admin PC successfully logged into MLS1 using kfpadmin through SSH

Finance to RADIUS Deny Test

```
C:\>ping 10.20.10.101

Pinging 10.20.10.101 with 32 bytes of data:

Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.

Ping statistics for 10.20.10.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Result: Pass - Finance traffic to RADIUS server 10.20.40.101 was successfully blocked.

Finance to IT Deny Test

```
C:\>ping 10.20.10.101

Pinging 10.20.10.101 with 32 bytes of data:

Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.
Reply from 10.20.20.1: Destination host unreachable.

Ping statistics for 10.20.10.101:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Result: Pass - Finance traffic to IT/Admin VLAN was successfully blocked.

ACL Verification

```
KFP-MLS1#
KFP-MLS1#show access-lists
Standard IP access list SSH-MGMT-ONLY
    10 permit 10.20.10.0 0.0.0.255 (4 match(es))
    20 permit host 10.20.99.100
    30 deny any
Extended IP access list FINANCE-ACL
    10 deny ip 10.20.20.0 0.0.0.255 10.20.99.0 0.0.0.255
    20 permit udp any eq bootpc any eq bootps (18 match(es))
    30 deny ip 10.20.20.0 0.0.0.255 host 10.20.40.11 (4 match(es))
    40 deny ip 10.20.20.0 0.0.0.255 host 10.20.40.12
    50 deny ip 10.20.20.0 0.0.0.255 10.20.10.0 0.0.0.255 (4 match(es))
    60 permit ip 10.20.20.0 0.0.0.255 any (5 match(es))

KFP-MLS1#
```

Result: Pass - ACL counters confirmed the permit and all deny rules matched the test results.